PAPER_1318 — Glueball Mass m_0++

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open (CLOSED) **Date:** June 11, 2026 **Calculator surface:** `calculate_paradox({"paradox": "glueball_mass"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1318})`

Master Expression

```
m_0++ = 2 × D_phys × Λ_QCD = 8 × 0.217 = 1.736 GeV
```

Match: 2.1% vs lattice 1.7 GeV

Interpretation

Integer-primitive identity: m_0++ = 2D_phys × Λ_QCD where 8 = 2×4 is the universal gluon-pair factor.

UQFF Locked Primitives

- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84, Λ = 0.00729735
- Λ_QCD = 0.217 GeV, v_Higgs = 246 GeV (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.
